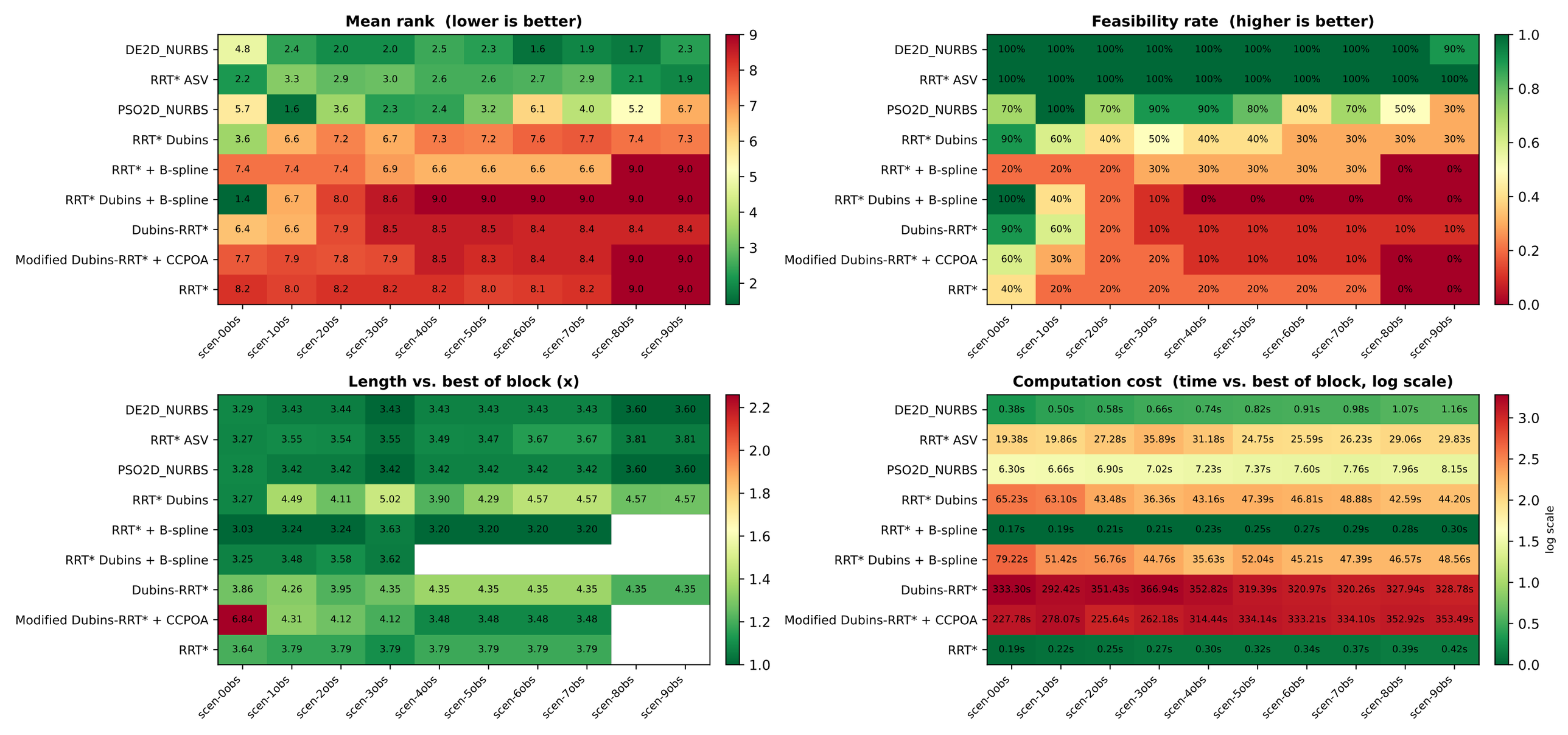

Mean rank (lower is better)
DE2D_NURBS 4.8 2.4 2.0 2.0 2.5 2.3 1.6 1.9 1.7 2.3
RRT* ASV 2.2 3.3 2.9 3.0 2.6 2.6 2.7 2.9 2.1 1.9
PSO2D_NURBS 5.7 1.6 3.6 2.3 2.4 3.2 6.1 4.0 5.2 6.7
RRT* Dubins 3.6 6.6 7.2 6.7 7.3 7.2 7.6 7.7 7.4 7.3
RRT* + B-spline 7.4 7.4 7.4 6.9 6.6 6.6 6.6 6.6 9.0 9.0
RRT* Dubins + B-spline 1.4 6.7 8.0 8.6 9.0 9.0 9.0 9.0 9.0 9.0
Dubins-RRT* 6.4 6.6 7.9 8.5 8.5 8.5 8.4 8.4 8.4 8.4
Modified Dubins-RRT* + CCPOA 7.7 7.9 7.8 7.9 8.5 8.3 8.4 8.4 9.0 9.0
RRT* 8.2 8.0 8.2 8.2 8.2 8.0 8.1 8.2 9.0 9.0
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
9 8 7 6 5 4 3 2
Feasibility rate (higher is better)
DE2D_NURBS 100% 100% 100% 100% 100% 100% 100% 100% 100% 90%
RRT* ASV 100% 100% 100% 100% 100% 100% 100% 100% 100% 100%
PSO2D_NURBS 70% 100% 70% 90% 90% 80% 40% 70% 50% 30%
RRT* Dubins 90% 60% 40% 50% 40% 40% 30% 30% 30% 30%
RRT* + B-spline 20% 20% 20% 30% 30% 30% 30% 30% 0% 0%
RRT* Dubins + B-spline 100% 40% 20% 10% 0% 0% 0% 0% 0% 0%
Dubins-RRT* 90% 60% 20% 10% 10% 10% 10% 10% 10% 10%
Modified Dubins-RRT* + CCPOA 60% 30% 20% 20% 10% 10% 10% 10% 0% 0%
RRT* 40% 20% 20% 20% 20% 20% 20% 20% 0% 0%
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
1.0 0.8 0.6 0.4 0.2 0.0
Length vs. best of block (x)
DE2D_NURBS 3.29 3.43 3.44 3.43 3.43 3.43 3.43 3.43 3.60 3.60
RRT* ASV 3.27 3.55 3.54 3.55 3.49 3.47 3.67 3.67 3.81 3.81
PSO2D_NURBS 3.28 3.42 3.42 3.42 3.42 3.42 3.42 3.42 3.60 3.60
RRT* Dubins 3.27 4.49 4.11 5.02 3.90 4.29 4.57 4.57 4.57 4.57
RRT* + B-spline 3.03 3.24 3.24 3.63 3.20 3.20 3.20 3.20
RRT* Dubins + B-spline 3.25 3.48 3.58 3.62
Dubins-RRT* 3.86 4.26 3.95 4.35 4.35 4.35 4.35 4.35 4.35 4.35
Modified Dubins-RRT* + CCPOA 6.84 4.31 4.12 4.12 3.48 3.48 3.48 3.48
RRT* 3.64 3.79 3.79 3.79 3.79 3.79 3.79 3.79
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
2.2 2.0 1.8 1.6 1.4 1.2 1.0
Computation cost (time vs. best of block, log scale)
DE2D_NURBS 0.38s 0.50s 0.58s 0.66s 0.74s 0.82s 0.91s 0.98s 1.07s 1.16s
RRT* ASV 19.38s 19.86s 27.28s 35.89s 31.18s 24.75s 25.59s 26.23s 29.06s 29.83s
PSO2D_NURBS 6.30s 6.66s 6.90s 7.02s 7.23s 7.37s 7.60s 7.76s 7.96s 8.15s
RRT* Dubins 65.23s 63.10s 43.48s 36.36s 43.16s 47.39s 46.81s 48.88s 42.59s 44.20s
RRT* + B-spline 0.17s 0.19s 0.21s 0.21s 0.23s 0.25s 0.27s 0.29s 0.28s 0.30s
RRT* Dubins + B-spline 79.22s 51.42s 56.76s 44.76s 35.63s 52.04s 45.21s 47.39s 46.57s 48.56s
Dubins-RRT* 333.30s 292.42s 351.43s 366.94s 352.82s 319.39s 320.97s 320.26s 327.94s 328.78s
Modified Dubins-RRT* + CCPOA 227.78s 278.07s 225.64s 262.18s 314.44s 334.14s 333.21s 334.10s 352.92s 353.49s
RRT* 0.19s 0.22s 0.25s 0.27s 0.30s 0.32s 0.34s 0.37s 0.39s 0.42s
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
log scale
3.0 2.5 2.0 1.5 1.0 0.5 0.0